

Multiplying Rational Numbers

Jonathan R. Bacolod

To multiply two **fractions**, multiply the numerators and multiply the denominators.

Cross-Canceling

Before multiplying, simplify by **cross-canceling** common factors whenever possible.

$$\left(\frac{2}{3}\right) \left(\frac{9}{4}\right) = \left(\frac{1}{3}\right) \left(\frac{9}{2}\right) = \frac{3}{2}$$

Multiplicative Identity

Any rational number multiplied by 1 remains unchanged.

This is called the **multiplicative identity**.

$$\frac{5}{7} \times 1 = \frac{5}{7}$$

Multiplying by Zero

Any rational number multiplied by 0 equals 0.

$$\frac{7}{9} \times 0 = 0$$

Mixed Numbers

Mixed numbers should be converted to **improper fractions** before multiplying.

$$2\frac{1}{2} = \frac{5}{2}$$

$$2\frac{1}{2} \times \frac{4}{5} = \frac{5}{2} \times \frac{4}{5} = 2$$

Simplest Form

Always simplify the final answer to its **lowest terms**.

$$\frac{8}{15} \times \frac{25}{16} = \frac{5}{6}$$

Examples

Example 1

Find the product.

$$\begin{pmatrix} 2 \\ -3 \end{pmatrix} \begin{pmatrix} 5 \\ 7 \end{pmatrix} = \frac{10}{21}$$

Example 2

Find the product.

$$\left(\frac{4}{5}\right) \left(\frac{3}{8}\right) = \frac{12}{40} = \frac{3}{10}$$

Example 3

Find the product.

$$2\frac{1}{2} \times \frac{4}{5} = \frac{5}{2} \times \frac{4}{5} = 2$$

Practice Exercises

Practice 1

Find the product. Express your answer in simplest form.

$$\begin{pmatrix} 2 \\ -3 \end{pmatrix} \begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

Practice 2

Find the product. Express your answer in simplest form.

$$\begin{pmatrix} 4 \\ -5 \end{pmatrix} \begin{pmatrix} 3 \\ -8 \end{pmatrix}$$

Practice 3

Find the product. Express your answer in simplest form.

$$\left(\frac{7}{9}\right) \left(\frac{6}{11}\right)$$

Practice 4

Find the product. Express your answer in simplest form.

$$\left(\frac{3}{4}\right) (12)$$

Practice 5

Find the product. Express your answer in simplest form.

$$\left(\frac{5}{6}\right) (18)$$

Practice 6

Find the product. Express your answer in simplest form.

$$2\frac{1}{2} \times \frac{4}{5}$$

Practice 7

Find the product. Express your answer in simplest form.

$$2\frac{3}{4} \times \frac{2}{7}$$

Practice 8

Find the product. Express your answer in simplest form.

$$\left(\frac{8}{15}\right) \left(\frac{25}{16}\right)$$

Practice 9

Find the product. Express your answer in simplest form.

$$\left(\frac{9}{14}\right) \left(\frac{7}{12}\right)$$

Practice 10

Find the product. Express your answer in simplest form.

$$\left(2\frac{1}{3}\right) \left(4\frac{1}{2}\right)$$

Activities

Activity 1

Find the product. Express your answer in simplest form.

$$\left(\frac{4}{9}\right) \left(\frac{3}{5}\right)$$

Activity 2

Find the product. Express your answer in simplest form.

$$\left(\frac{7}{10}\right) \left(\frac{5}{14}\right)$$

Activity 3

Find the product. Express your answer in simplest form.

$$\left(\frac{8}{13}\right) \left(\frac{3}{4}\right)$$

Activity 4

Find the product. Express your answer in simplest form.

$$\left(\frac{5}{6}\right) (18)$$

Activity 5

Find the product. Express your answer in simplest form.

$$\left(\frac{7}{8}\right)(16)$$

Activity 6

Find the product. Express your answer in simplest form.

$$3\frac{2}{3} \times \frac{9}{10}$$

Activity 7

Find the product. Express your answer in simplest form.

$$2\frac{1}{4} \times \frac{4}{9}$$

Activity 8

Find the product. Express your answer in simplest form.

$$\left(\frac{12}{35}\right) \left(\frac{14}{9}\right)$$

Activity 9

Find the product. Express your answer in simplest form.

$$\left(\frac{15}{22}\right) \left(\frac{11}{20}\right)$$

Activity 10

Find the product. Express your answer in simplest form.

$$\left(3\frac{1}{5}\right) \left(2\frac{1}{4}\right)$$

Thank you!